

Precision Guidance System

- By Team Risky 5

Team Details

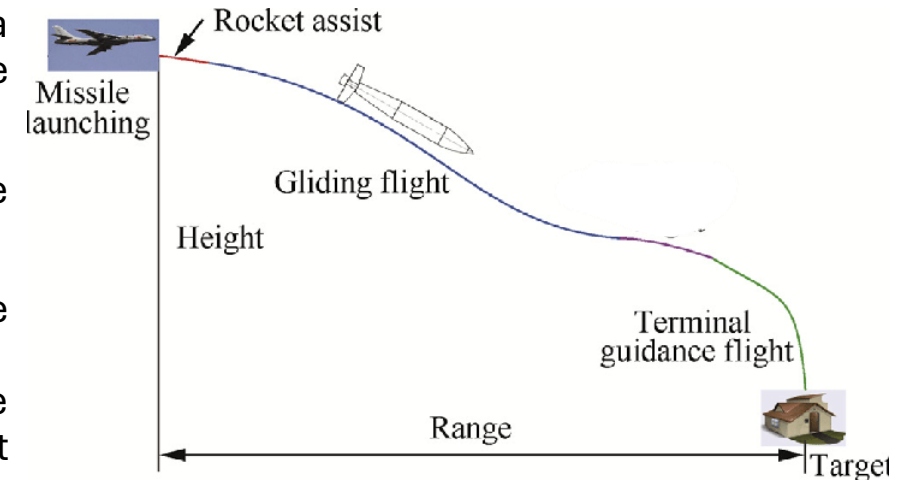
- **Team Name:** Risky 5
- **Startup Name:** InfraCore Systems LLP
- **Name of Team Lead:** Kalrav Mathur
- **Name of the Institution:** Shiv Nadar University
- **Name of the processor:** SHAKTI Yamuna
- **Name of the Mentor:** Dr. Venkatnarayan Hariharan

Precision Guidance System

Target Application: Security and Surveillance

Objectives of the proposal:

- To develop a terminal-phase Imaging Infrared (IIR) seeker prototype for a precision guided missile system on a fully indigenous hardware stack using the SHAKTI processor.
- Running the main logic on the SHAKTI Yamuna and offloading compute intensive tasks of image correlation algorithms to FPGA for parallel execution.
- To deliver a physical prototype that demonstrates the complete guidance functionality.
This system will lock onto a physical target, visually track it by matching live thermal images with preloaded satellite imagery, and output the exact guidance commands.

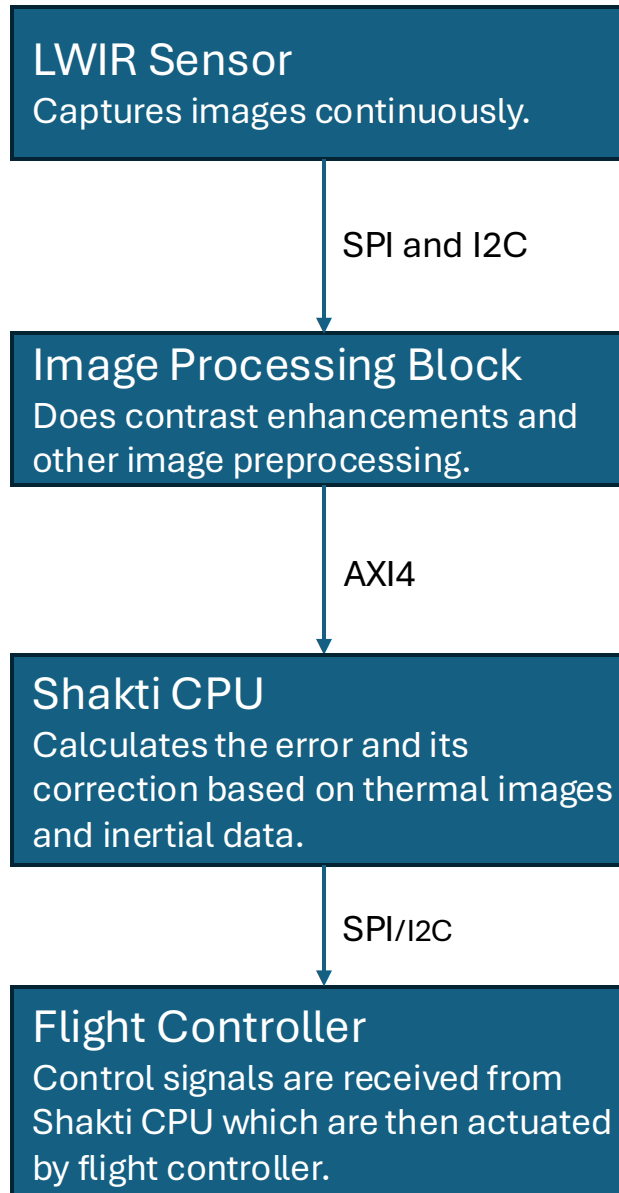


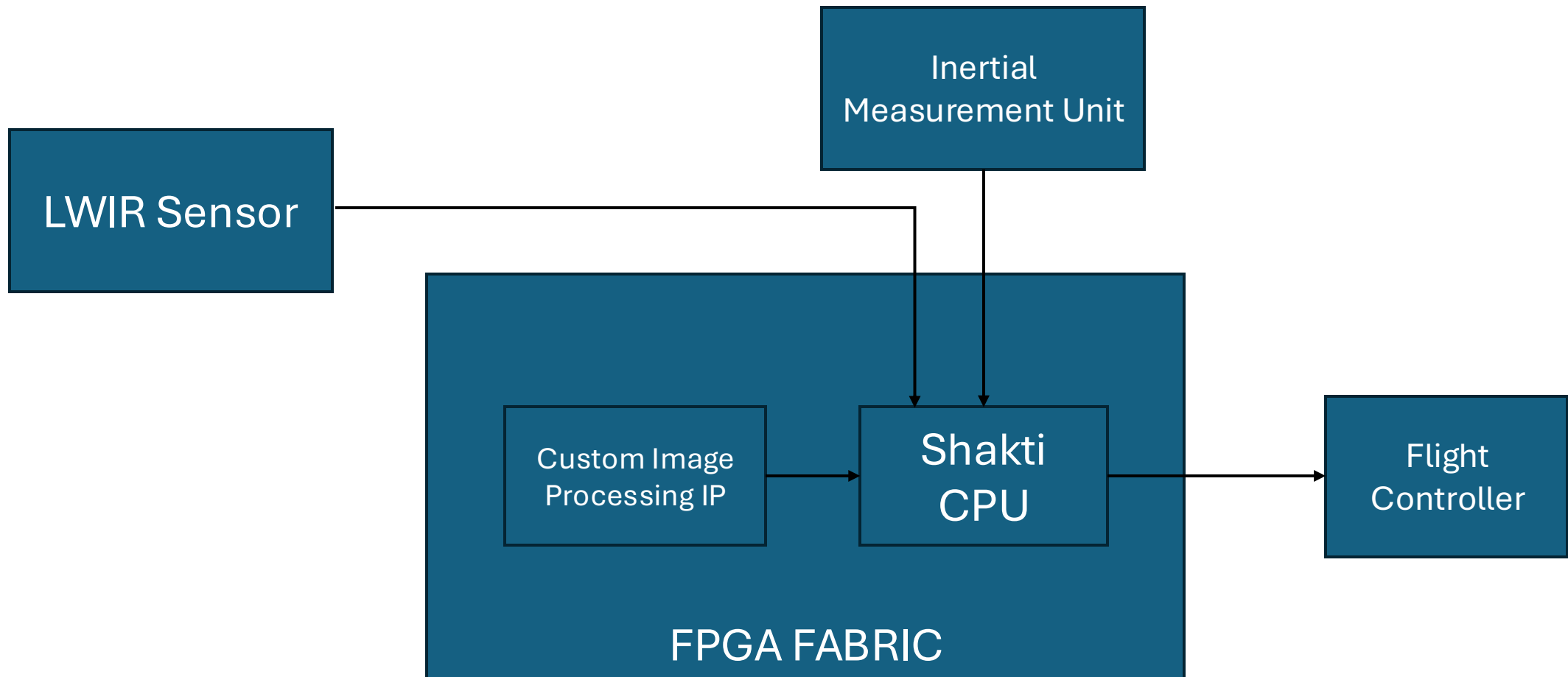
Novelty

- The project addresses a critical capability gap for India's strategic autonomy by replacing imported IIR-guided systems like SCALP-EG and HAMMER, and lays the foundation for future R&D in seeker design.
- Its core novelty is the implementation of a terminal-phase IIR seeker emulator on fully indigenous hardware, eliminating reliance on foreign black-box CPUs/GPUs and the security risks they pose.
- Rather than simply running software on a processor, we use a hybrid architecture where the FPGA handles parallel pixel-level tasks, freeing the SHAKTI core for high-level processing — enabling real-time performance.
- Unlike proprietary military systems, our emulator is modular and adaptable, allowing repurposing for civilian and security applications like drone navigation, border surveillance, precision agriculture, and disaster relief.
- The project provides a low-cost, indigenous compute platform for thermal-guided systems, enabling scalable deployment by Indian MSMEs under Make in India and iDEX (Innovations for Defence Excellence) , ideal for disposable military use.

Proposed Approach

- Our goal is to guide a "**Fire & Forget**" ballistic missile in its terminal phase (after cruising stage to blast) towards its target.
- We take inputs from a camera (in Infra-Red spectrum) and use image-correlation algorithms to compute the shift in missile's heading.
- This can then be translated into the control signals that govern the missile's heading (yaw, pitch, roll, AoA (Angle of Attack), etc.).
- We get additional information about heading from the inertial sensors mentioned above.





Block Diagram

Project Milestones Overview

Milestone 1: MVP (Due 24th Oct 2025)

Goal: Basic Image Capture & Processing Pipeline

- Environment setup for Artix-7 FPGA & SHAKTI RISC-V
- Interface LWIR sensor to capture single thermal frame
- Verilog logic for frame transfer to SHAKTI soft-core
- C/C++ matching algorithm detects predefined target

Deliverables:

- Live image via SHAKTI serial output
- “MATCH FOUND” result + coordinates

Milestone 2: PoC (Due 25th Dec 2025)

Goal: Real-time Video Processing Pipeline

- Continuous thermal video stream(>10 FPS)
- FPGA IP: contrast, filtering, edge detection
- Real-time target detection (NCC) +IMU interfacing

Deliverables:

- Processed video on monitor
- LED indicator for “TARGET LOCKED”
- Serial output: error vector + IMU data
- Source code (FPGA + SHAKTI)

Milestone 3: Final Demo (Due 20th Apr 2026)

Goal: Full Guidance System with Live Target Tracking

- Image + IMU fused for control commands
- Robustness to occlusions & reacquisition
- Real-time bounding box + command output

Deliverables:

- Working prototype on test stand
- Live demo with target lock & recovery
- Guidance commands (e.g., YAW_RIGHT: 3.2°)
- Final report + code + demo video

Business Potential

- **Large and Growing Global PGM Market**

The global PGM market is valued at USD 35–38 billion today, rising to USD 50–55 billion by 2030, with consistent CAGR around 5–6%

- **Substantial Indian Defence Funding**

India's defence budget is ₹6.81 trillion (~USD 78 B) for FY 2025–26, with ₹1.8 trillion (~26%) dedicated to modernization and procurement

- **Strong Push for Domestic Procurement**

The government has earmarked 75% of modernisation capital (~₹1.11 lakh crore) exclusively for domestic sourcing, reinforcing self-reliance goals

- **Export Potential via ITAR-Free, Sensor-Agnostic Design**

As defense exports rise toward USD 6 billion by 2029, an indigenous compute-and-guidance platform built on open Indian standards unlocks opportunities in friendly markets across Asia, Africa, and Latin America

Implementation strategy

Target Market / Initial Application

The solution is aimed at key stakeholders in the Indian defence ecosystem, including DRDO labs, the armed forces, and private defense Manufacturers. The initial application focuses on expendable compute modules for UAVs, loitering munitions, and training munitions, where cost-efficiency is critical.

Go-to-Market Approach

We plan to engage with DRDO and BEL through initiatives like iDEX and TDF to support early-stage development and validation, while partnerships with Indian MSMEs will enable manufacturing and field deployment, positioning the product as a cost-effective alternative to imported compute boards.

IP & Licensing

Our architecture is modular and reusable — it can be licensed to defense manufacturers or spun off into a startup. We're also open to joint development and integration with existing seeker/guidance platforms.

Prototype & Pilot Phase

We'll prototype the system using SHAKTI/VEGA processors, followed by testing in simulated UAV or missile environments to demonstrate real-time performance and robustness.

Scaling Strategy

Once validated, we plan to scale through MSME manufacturing under Make in India. In the future, the same platform can be adapted for civilian drone-based applications such as disaster response, border monitoring, and surveillance.

Review Meeting 2

20 December 2025

Action Points discussed in last meeting

- **Communication Protocols:**

- Flight Controller Module
- LWIR Camera

- **LUT Usage Optimization:**

- Image Processing IP LUT Utilization
- Shakti SOC Peripheral Optimization

- **Code Showcase:**

- Shakti Embedded Application
- Image Processing IP

Progress Made since last meeting

Custom Image Processing IP:

- Implementation of NCC Algorithm without OpenCV in C++.
- Conversion of C++ code to Verilog. (Manually)
- Functional verification and debugging the Verilog Module.
- Optimization on 2 grounds i.e. LUT usage and Parallel Execution.

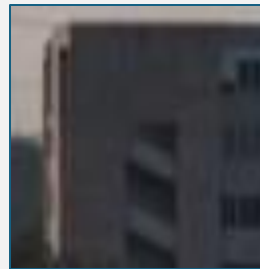
Name	Slice LUTs (63400)	Block RAM Tile (135)	DSPs (240)	Bonded IOB (210)	BUFGCTRL (32)	MMCME2_ADV (6)	Slice Registers (126800)	F7 Muxes (31700)	Slice (15850)	LUT as Logic (63400)
top_ncc_7seg	11118	33	208	20	2	1	1802	4	3647	11118
disp_inst (seven_seg_driver)	5	0	0	0	0	0		0	9	5
ncc_core (ncc_rtl)	11079	33	208	0	0	0		4	3639	11079
u_clkgen (clk_wiz_0)	0	0	0	0	2	1		0	0	0

Progress Made since last meeting

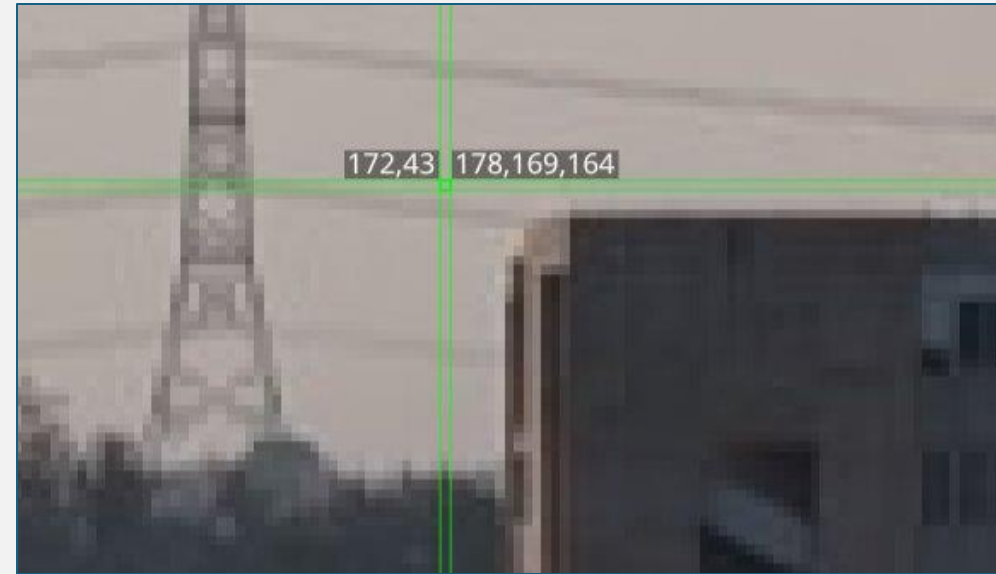
Custom Image Processing IP: Test Inputs



298 x 224 Image



64x64 Template



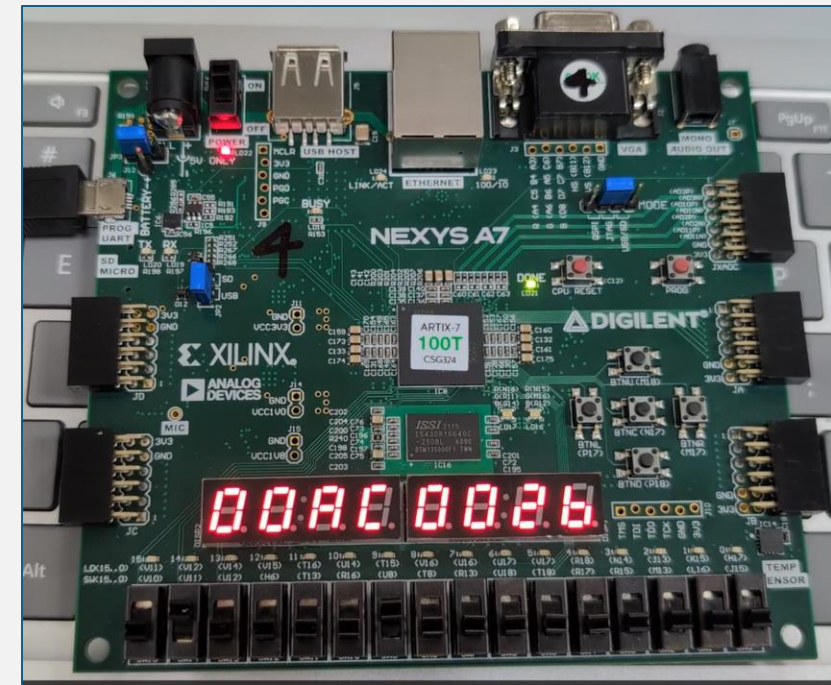
(172,43) Top Left Coordinate

Progress Made since last meeting

Custom Image Processing IP: Results

```
// Part 10
t=0 ns start=0 reset=1 busy=x done=x x=x y=x
t=1000 ns start=0 reset=1 busy=0 done=0 x=0 y=0
t=5000 ns start=0 reset=0 busy=0 done=0 x=0 y=0
t=7000 ns start=1 reset=0 busy=1 done=0 x=0 y=0
t=8000 ns start=0 reset=0 busy=1 done=0 x=0 y=0
t=155203437000 ns start=0 reset=0 busy=0 done=1 x=172 y=43
t=155203438000 ns start=0 reset=0 busy=0 done=0 x=172 y=43
```

Behavioral Simulation



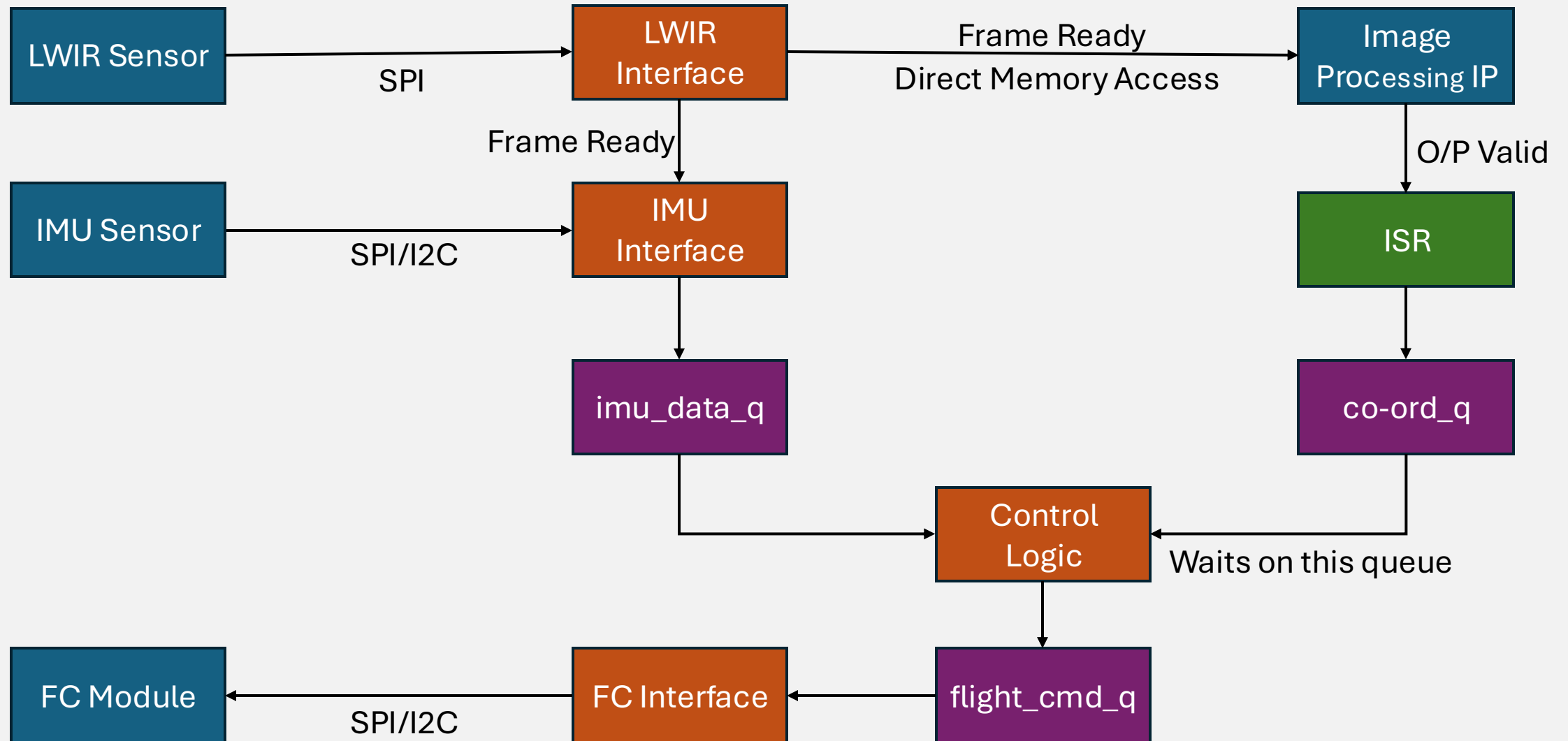
Result display using 7 segment display on
Artix-7 FPGA

Progress Made since last meeting

Shakti Embedded Application:

- Understanding and getting familiar with Shakti SDK.
- Initial FreeRTOS application development with interrupts, other required peripherals and GPIOs.
- Architecture design of the application.
- Tasks implementation and Inter-Process communication setup.
- Core logic implementation with PID controller for correction.

RTOS Application Architecture



Progress Made since last meeting

Shakti Custom Peripheral SOC:

- Required Peripherals

- 1 UART (Serial monitor & debug)
- 1 SPI (camera interface/data)
- 1 I2C (camera control commands)

- Status

- Removal of unused peripherals is in progress
- Debugging build issues arising from peripheral removal in the modified SoC

Progress Made since last meeting

Component Research:

Selection

- **LWIR Sensor:** Selected **Waveshare Thermal Camera HAT** for R&D. Features 80×62 resolution at 25 FPS.
- **Flight Controller & IMU:** Selected **Radiolink F722**. Eliminates the need for a separate IMU by leveraging the onboard ICM42688 Gyroscope and BMP280 Barometer.

Interfacing Protocols

- **LWIR Sensor:** Utilizes a dual-protocol interface; I2C for configuring camera registers and SPI for thermal video data.
- **Flight Controller:** Standardized communication (UART/SBUS) to receive guidance commands from the Shakti SoC.

Startup Formation Status

- ✓ Certificate of Incorporation
- ✓ LLP Agreement
- ✓ PAN of LLP
- ✓ Company Bank Account
- ✓ DPIIT Registration Certificate

Proposed Plan

- Finish development of optimized and balanced Image Processing IP.
- Complete the startup formalities.
- Order the required Hardware Modules. (LWIR sensor and Flight Controller).
- Interfacing and testing of the hardware modules.
- Further development for integrating everything in embedded application.

Review Meeting 3

24 January 2026

Progress Made since last meeting

Shakti Embedded Application:

- Implemented UART, SPI and I2C bare-metal helper functions.
- PID control loops for Pitch and Yaw error correction done.
- Calculation of correction commands is implemented.
- Interfacing development for CAM, IMU and Image IP is ongoing.
- Development NCC algorithm for target matching on Yamuna SOC is ongoing.

Progress Made since last meeting

Shakti Custom Peripheral SOC:

- Required Peripherals

- 2 UART (Serial monitor and IMU module)
- 1 SPI (interface with Camera Module)

- Status

- Critical peripherals required for our use case had been identified and necessary addition/removal has been done.

Progress Made since last meeting

Components:

- **Hardware Procured:**
 - Camera Module: Seeed Studios XIAO ESP32S3 Sense
 - Supports I2C, SPI and UART.
 - IMU Module: Adafruit BNO085
 - Supports I2C, SPI and UART.

Progress Made since last meeting

Custom Image Processing IP:

- Optimization is ongoing for making it faster and smaller.
- Interfacing with Shakti SOC is ongoing.

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Milestone 1: MVP

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Milestone 2: PoC

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Thank You

Precision Guidance System

MVP Presentation - Infracore Systems LLP (Team Risky 5)

27 February 2026

Team Details

- **Team Name:** Risky 5
- **Startup Name:** Infracore Systems LLP
- **Name of Team Lead:** Kalrav Mathur
- **Name of the Institution:** Shiv Nadar Institution of Eminence, Delhi NCR
- **Name of the processor:** Shakti Yamuna
- **Name of the Mentor:** Dr. Venkatnarayan Hariharan

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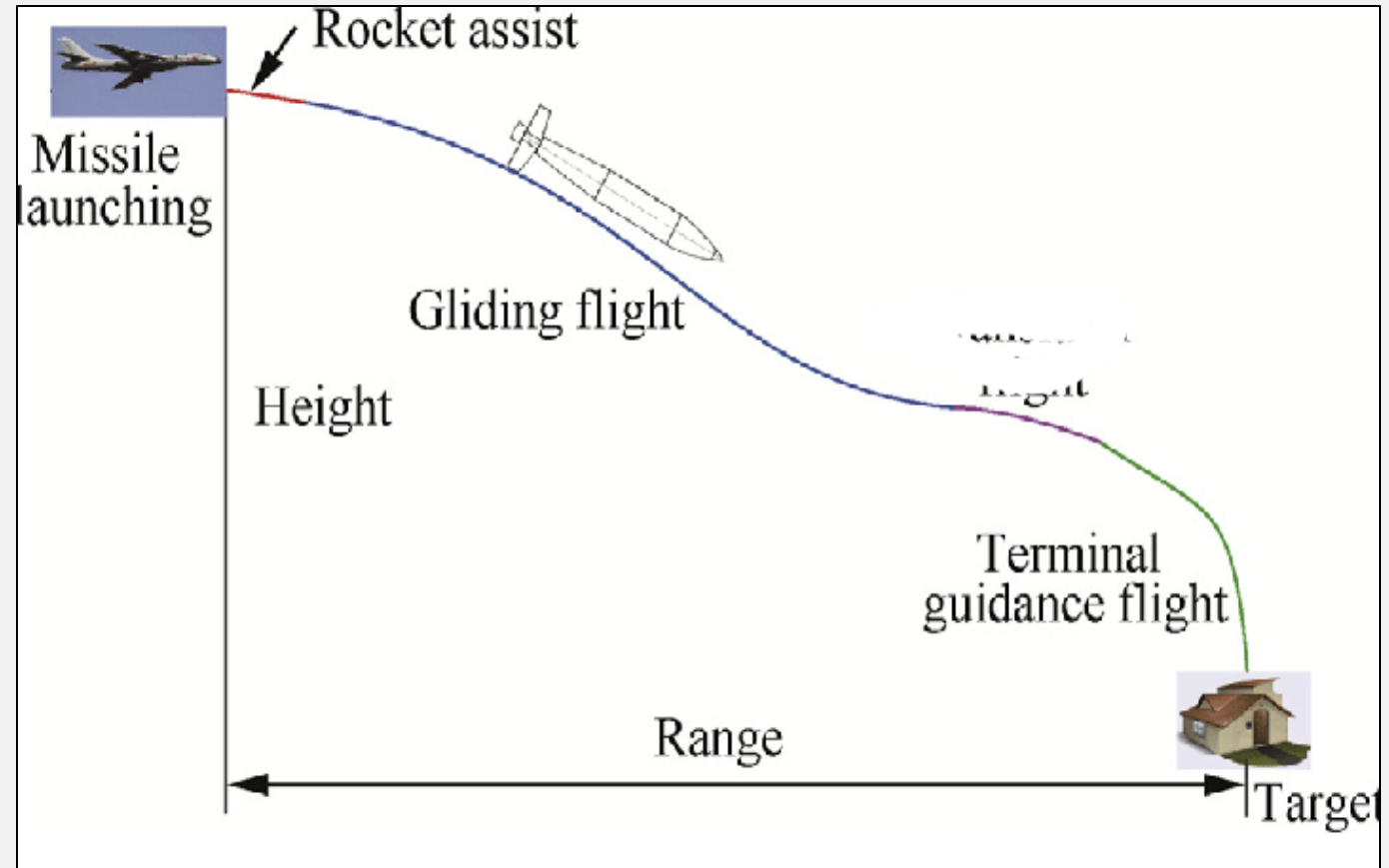
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Precision Guidance System

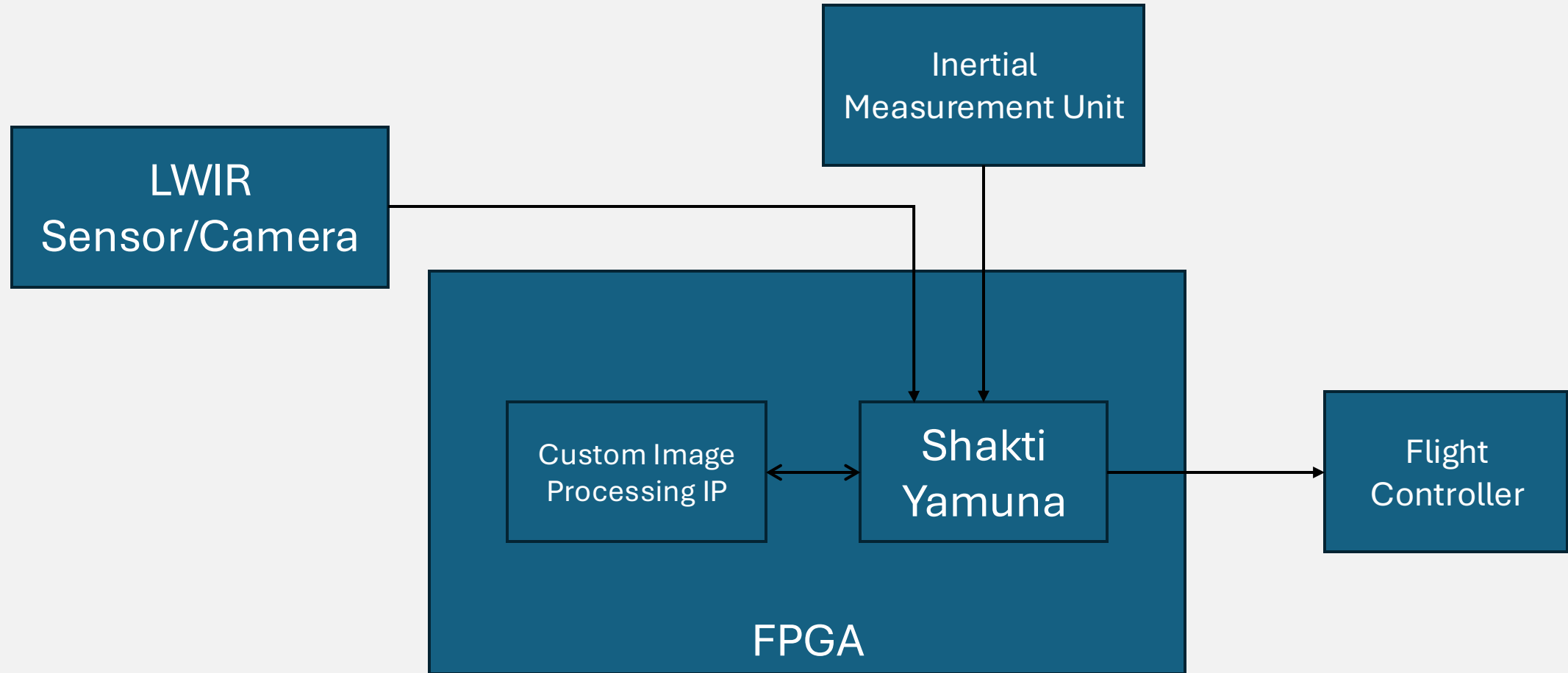
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Objectives of the proposal:

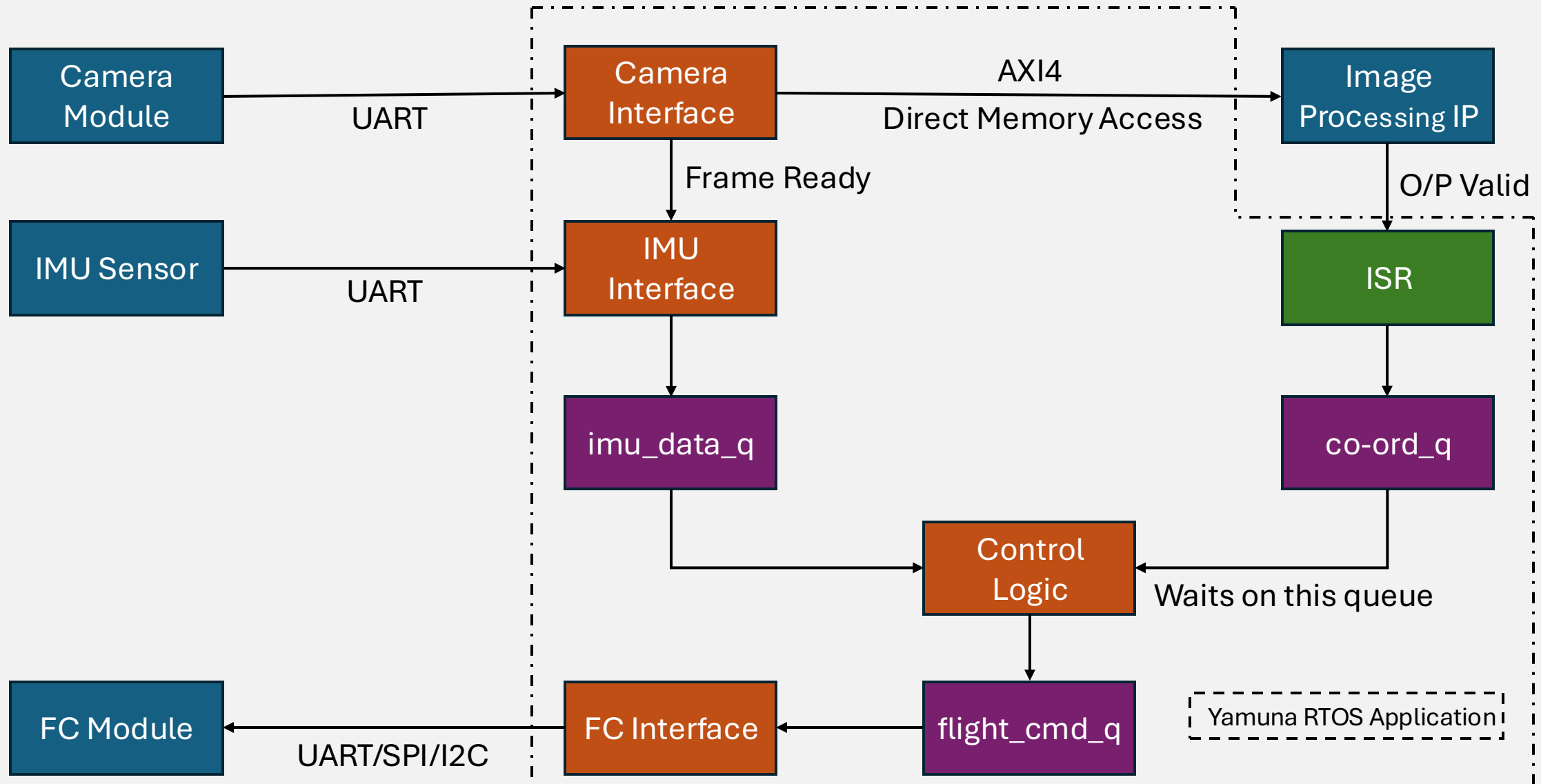
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Block Diagram



Yamuna Application Architecture



Shakti Custom Peripheral SOC

Required Peripherals

- 3 UART
 - Camera
 - IMU
 - Serial Monitor

Removed Peripherals

- 2 SPI Controllers
- 2 I2C Controllers

LUTs Utilization - Before Optimization v/s After

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	50371	0	0	63400	79.45
LUT as Logic	49189	0	0	63400	77.59
LUT as Memory	1182	0	0	19000	6.22
LUT as Distributed RAM	958	0			
LUT as Shift Register	224	0			

Default configuration – LUTs

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	43690	0	0	63400	68.91
LUT as Logic	42512	0	0	63400	67.05
LUT as Memory	1178	0	0	19000	6.20
LUT as Distributed RAM	958	0			
LUT as Shift Register	220	0			

Optimized configuration - LUTs

After removing the unused peripherals, the LUTs utilization was reduced by ~10.54% (~13.26% relative reduction).

Hardware Procured

- **Camera Module:** Seeed Studios XIAO ESP32S3 Sense
- **Inertial Measurement Unit:** Adafruit BNO085 IMU
- Breadboard, jumper wires and Qwiic Connectors.

Interfacing Protocols

- **Camera:** UART
- **IMU:** UART

Camera Interface

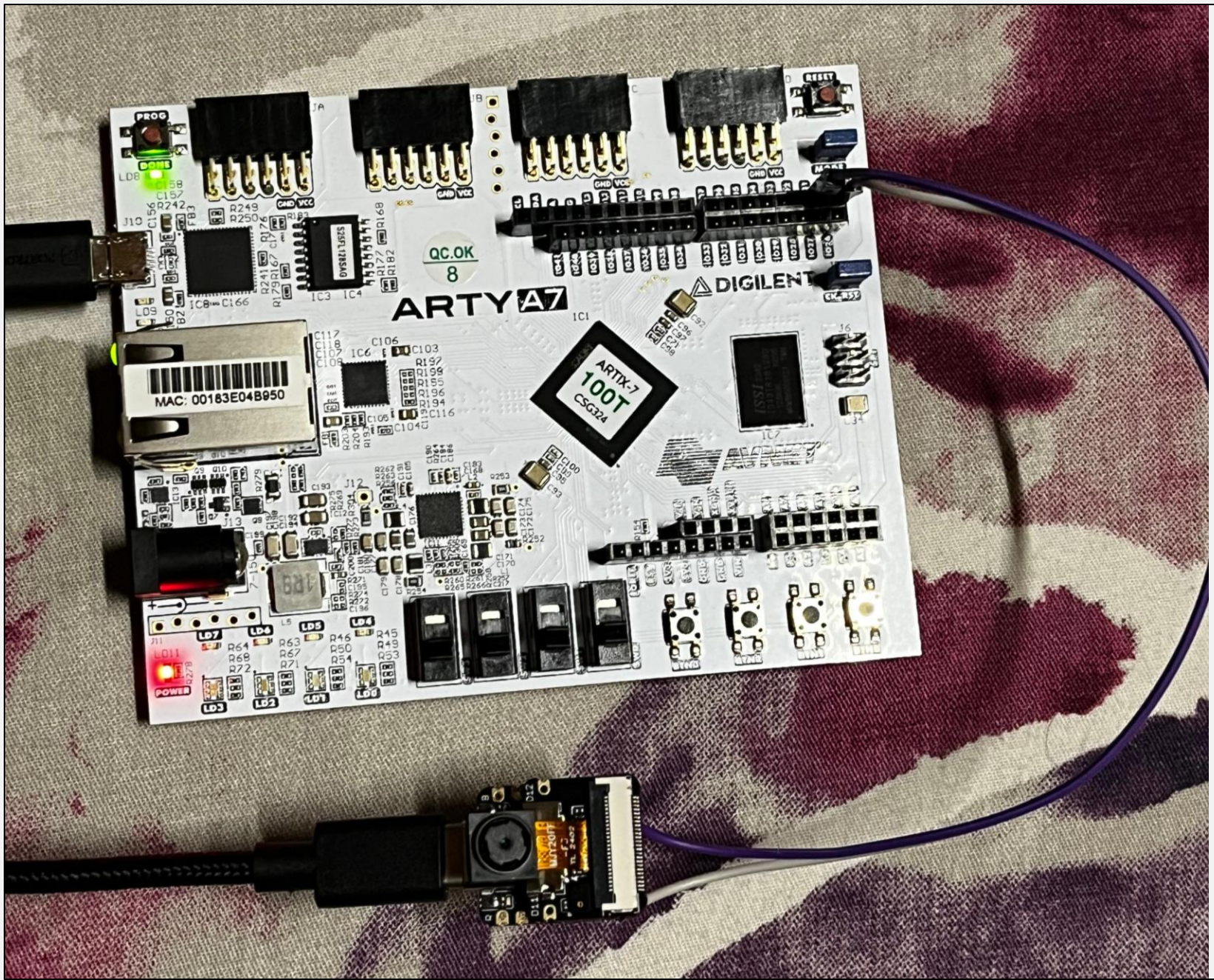
- UART interface between ESP32S3 cam and Shakti Yamuna.
- 153600 baud rate, 2 stop bits at ESP (1 at Yamuna). No RTS/CTS.

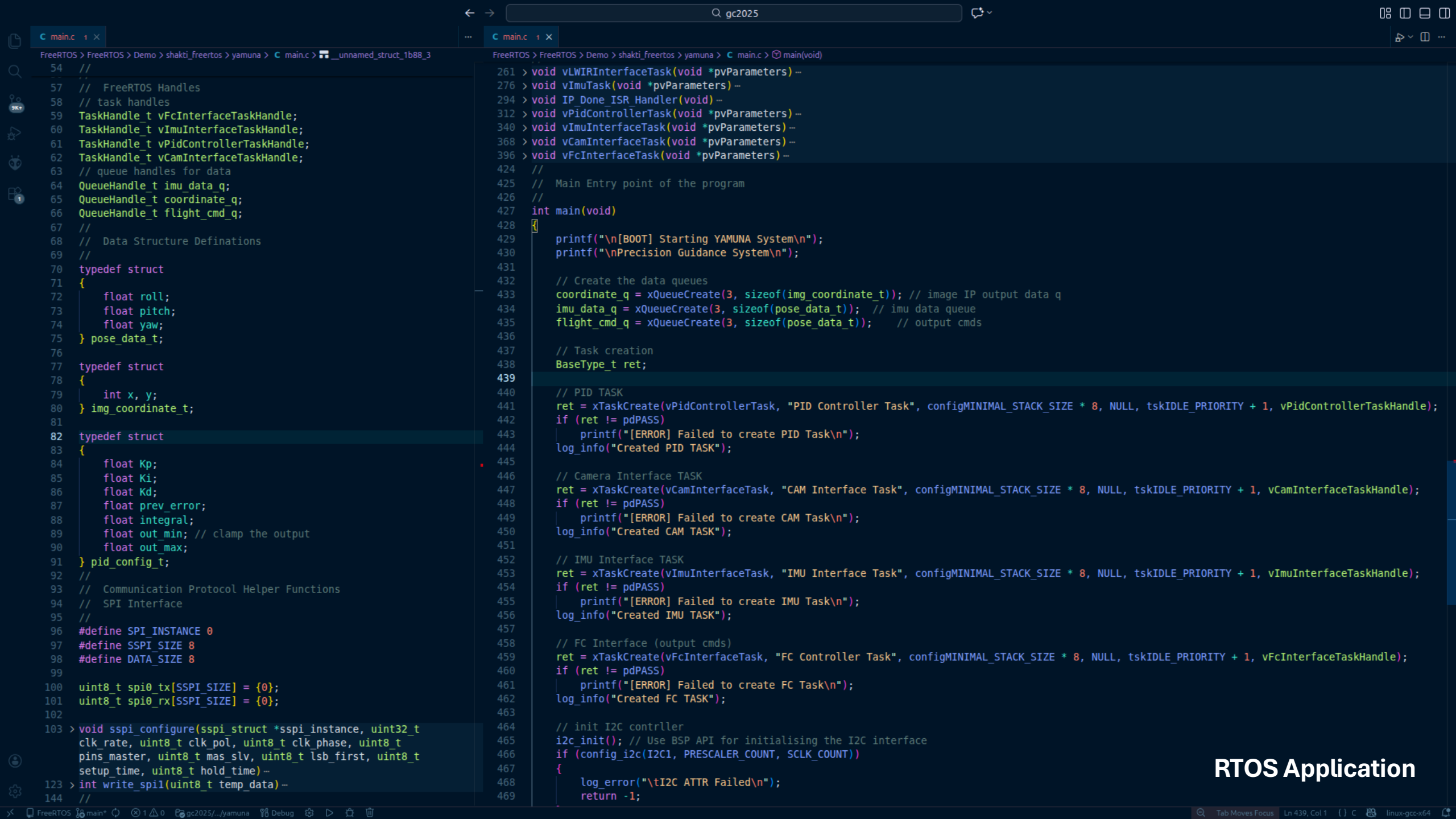
```
Monitor Mode Serial View Mode Text Port /dev/ttyACM0 - Espressif
05:15:55:603 ->
05:15:55:603 -> [LOOP] Captured Frame Len: 1988
05:15:55:603 -> Sending Frame
05:16:19:459 -> [LOOP] Frame sent (1988 bytes)
05:16:24:459 ->
05:16:24:459 ->
05:16:24:459 -> [LOOP] Captured Frame Len: 1985
05:16:24:459 -> Sending Frame
05:16:48:279 -> [LOOP] Frame sent (1985 bytes)
05:16:53:278 ->
05:16:53:278 ->
05:16:53:278 -> [LOOP] Captured Frame Len: 2009
05:16:53:278 -> Sending Frame
05:17:17:386 -> [LOOP] Frame sent (2009 bytes)
|
```

Frame bytes being sent from ESP32S3

```
GTKTerm - /dev/ttyUSB1 19200-8-N-1
File Edit Log Configuration Control signals View Help
[20490.23h.45m.50s.696] Frame RECVD 1984 bytes
[20490.23h.46m.19s.543] Frame RECVD 1988 bytes
[20490.23h.46m.48s.358] Frame RECVD 1985 bytes
[20490.23h.47m.17s.476] Frame RECVD 2010 bytes
```

Frame bytes being received at Yamuna





NCC Algorithm Implementation

```
File Edit Log Configuration Controlsignals View Help

[20490.19h.07m.57s.161] inside main
[20490.19h.07m.57s.177] Starting NCC compute
[20490.19h.07m.58s.745]
                        [20490.19h.07m.58s.745] Target Box Top left
X
[20490.19h.07m.58s.761] X: 62   Y: 28
[20490.19h.07m.58s.761] NCC Score: 0.842782
```

NCC Compute with pre-stored images

```
GTKTerm - /dev/ttyUSB1 115200-8-N-1
File Edit Log Configuration Controlsignals View Help

[20503.18h.32m.10s.355] [MAIN][LOOP][INFO] getting frame
[20503.18h.32m.17s.219] [MAIN][GET_FRAME][INFO] Frame RCVD: 19200 Bytes
[20503.18h.32m.17s.219] [MAIN][LOOP][INFO] starting ncc compute
[20503.18h.32m.32s.531] [MAIN][LOOP][INFO] RESULT: X: 34      Y: 25      SCORE: 1.000031
[20503.18h.32m.32s.531]
[20503.18h.32m.32s.531]
[20503.18h.32m.32s.531]
[20503.18h.32m.33s.731] [MAIN][LOOP][INFO] getting frame
[20503.18h.32m.36s.339] [MAIN][GET_FRAME][INFO] Frame RCVD: 19200 Bytes
[20503.18h.32m.36s.339] [MAIN][LOOP][INFO] starting ncc compute
```

NCC Compute with live images

Custom Image Processing IP

- Implementation of Normalized Cross Correlation (NCC) Algorithm in Verilog RTL
- Functional Verification using Vivado testbench
- Implemented on Artix 7 FPGA

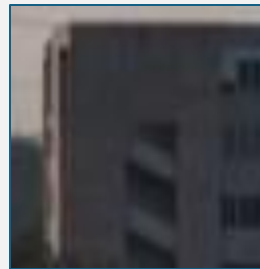
```
module ncc_rtl #(
    parameter integer W_IMG    = 298,
    parameter integer H_IMG    = 224,
    parameter integer W_TPL    = 64,
    parameter integer H_TPL    = 64,
    parameter      IMG_MEMFILE = "image.mem",
    parameter      TPL_MEMFILE = "target.mem"
) (
    input wire      clk,
    input wire      rst,           // active-high
    input wire      start,
    output reg      busy,
    output reg      done,
    output reg [31:0] out_x,
    output reg [31:0] out_y
);
```

Name	Slice LUTs (63400)	Block RAM Tile (135)	DSPs (240)	Bonded IOB (210)	BUFGCTRL (32)	MMCME2_ADV (6)	Slice Registers (126800)	F7 Muxes (31700)	Slice (15850)	LUT as Logic (63400)
top_ncc_7seg	11118	33	208	20	2	1	1802	4	3647	11118
disp_inst (seven_seg_driver)	5	0	0	0	0	0		0	9	5
ncc_core (ncc_rtl)	11079	33	208	0	0	0		4	3639	11079
u_clkgen (clk_wiz_0)	0	0	0	0	2	1		0	0	0

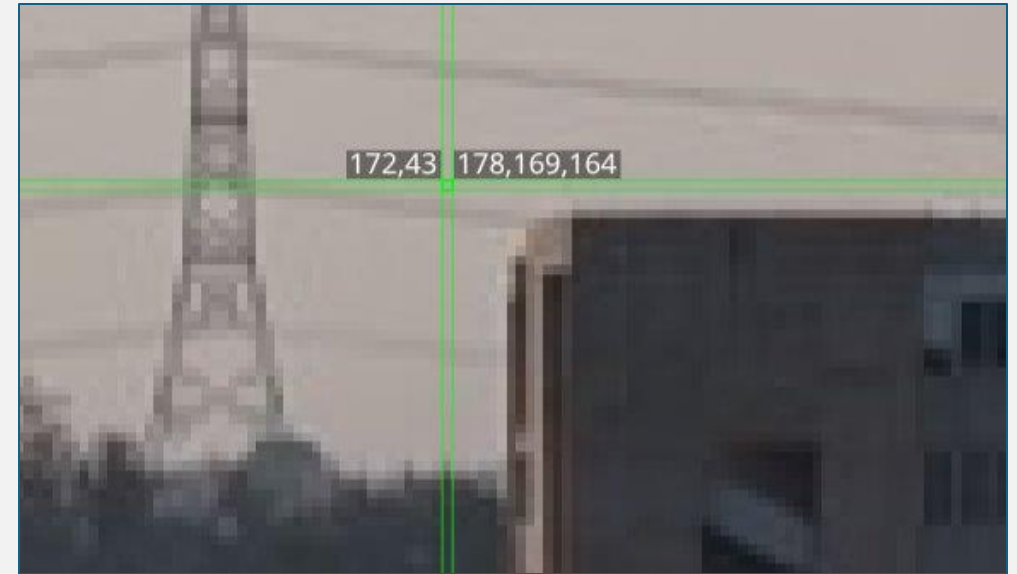
Custom Image Processing IP: Test Inputs



298 x 224 Image



64x64 Template



(172,43) Top Left Coordinate

Custom Image Processing IP: Results

```
// Part 19
t=0 ns start=0 reset=1 busy=x done=x x=x y=x
t=1000 ns start=0 reset=1 busy=0 done=0 x=0 y=0
t=5000 ns start=0 reset=0 busy=0 done=0 x=0 y=0
t=7000 ns start=1 reset=0 busy=1 done=0 x=0 y=0
t=8000 ns start=0 reset=0 busy=1 done=0 x=0 y=0
t=155203437000 ns start=0 reset=0 busy=0 done=1 x=172 y=43
t=155203438000 ns start=0 reset=0 busy=0 done=0 x=172 y=43
```

Behavioral Simulation

- Ongoing interfacing with Shakti core and optimization on 2 grounds: LUT usage and Parallel Execution

AXI Standalone IP For Testing

- Implementation of a simple AXI interfacing module.
- Functional Verification using Vivado testbench
- Working on creating a wrapper around our existing Verilog Image IP.
- Currently experimenting with AXI4-Lite and Axi Stream.

```
// AXI Master Interface (Connects to Slave)
output reg [3:0] m_axi_awaddr, output reg m_axi_awvalid, input wire m_axi_awready,
output reg [31:0] m_axi_wdata, output reg m_axi_wvalid, input wire m_axi_wready,
input wire m_axi_bvalid, output reg m_axi_bready,
output reg [3:0] m_axi_araddr, output reg m_axi_arvalid, input wire m_axi_arready,
input wire [31:0] m_axi_rdata, input wire m_axi_rvalid, output reg m_axi_rready
);

localparam S_IDLE = 0, S_WRITE = 1, S_WAIT_B = 2, S_READ = 3, S_WAIT_R = 4, S_CHECK = 5, S_STOP = 6;
reg [3:0] state;
```

AXI4-Lite Loopback Simulation

```
-----
[20000 ns] SYSTEM RESET RELEASED
[55000 ns] WRITE REQ  : Address 0x0
[55000 ns] WRITE DATA : Val =          10
[65000 ns] WRITE RESP : OKAY
[95000 ns] READ REQ   : Address 0x4
[105000 ns] READ DATA : Val =          11 (Expected 11)
SUCCESS: Sent 10, Received          11
-----
RESULT: [PASSED] Hardware Adder Works!
```

Market Need & Strategic Opportunity

- **India depends heavily on imported high-end guidance and vision systems**, which are expensive, restricted, and not fully customizable for Indian operational needs
- **Government and industry are actively looking for indigenous alternatives** under AtmaNirbhar Bharat, iDEX, and defence innovation programs

How the Startup Makes Money

- **Primary customers include government labs, defence PSUs, and Indian deep-tech startups** working on drones, surveillance systems, and autonomous platforms
- **Revenue is generated through paid pilots, licensing, customization contracts, and long-term support**, rather than one-time sales

Scaling into a Sustainable Indian Startup

- **Start with the Defence Application then expand to multiple civilian and dual-use sectors**, including security, infrastructure monitoring, and disaster response
- **Grow as an IP-driven company aligned with national priorities**, enabling exports to friendly countries and partnerships with Indian manufacturers

Startup Formation Status

- ✓ Certificate of Incorporation
- ✓ LLP Agreement
- ✓ PAN of LLP
- ✓ Company Bank Account
- ✓ DPIIT Registration Certificate

Thank You

Precision Guidance System

MVP Evaluation Feedback & Next Steps - Infracore Systems LLP
(Team Risky 5)

14th March 2026

Key Feedback from MVP Evaluation

- GPS-denied navigation is considered a largely solved problem, with existing approaches such as triangulation and other alternative navigation techniques already available.
- Since missiles or drones typically use GPS for long-range navigation, the value addition of our proposed system needs clearer justification.
- Existing sensing and guidance solutions were highlighted, and clear differentiation from current approaches was recommended.
- Standard IR sensors are commonly used for tracking moving targets with strong heat signatures; the choice of LWIR sensors in our design needs further justification.

Key Feedback from MVP Evaluation

- A human-in-the-loop approach was suggested, where live camera feeds could be transmitted to a ground station for manual guidance decisions.
- Further clarity on the novelty and practical contribution of the proposed system was recommended.
- We are currently in discussions with a domain expert to onboard them as a mentor/advisor and incorporate their guidance in refining the project direction.

Thank You

Infrared Search and Track System

POC Presentation - Infracore Systems LLP (Team Risky 5)

11 April 2026

POC Overview

- **Project Title:** Infrared Search and Track System
- **Team Name:** Infracore Systems LLP (Risky 5)
- **Members:** Akansh S., Akhil S., Dr. Venkatnarayan H., Kalrav M., Saksham M., Tushar J.
- **Problem Statement:** Active radar systems are highly vulnerable to electronic warfare (jamming/spoofing) and struggle to detect low-RCS (Radar Cross Section) threats.
- **Solution:** A passive Infrared Search and Track system. We utilize an FPGA for parallel spatial-temporal pattern matching to isolate targets, combined with a Shakti RISC-V processor to provide real-time I/O feedback to defence actuators.

Progress and Milestones

Achieved

- Interface sensor to capture single frame
- Verilog logic for frame transfer from SHAKTI soft-core
- NCC matching algorithm detects predefined target
- Live image via SHAKTI serial output
- “MATCH FOUND” result + coordinates
- KCF Fast pattern match algorithm implementation
- Architecture design for IRST
- Yamuna Embedded RTOS application to handle feedback
- Interface KCF with Yamuna using AXI and DMA (Ongoing)

Planned

- KCF Algorithm port to Verilog
- Complete the detection and notification loop
- Benchmark and compare response time
- Optimize the latency and delays in the loop

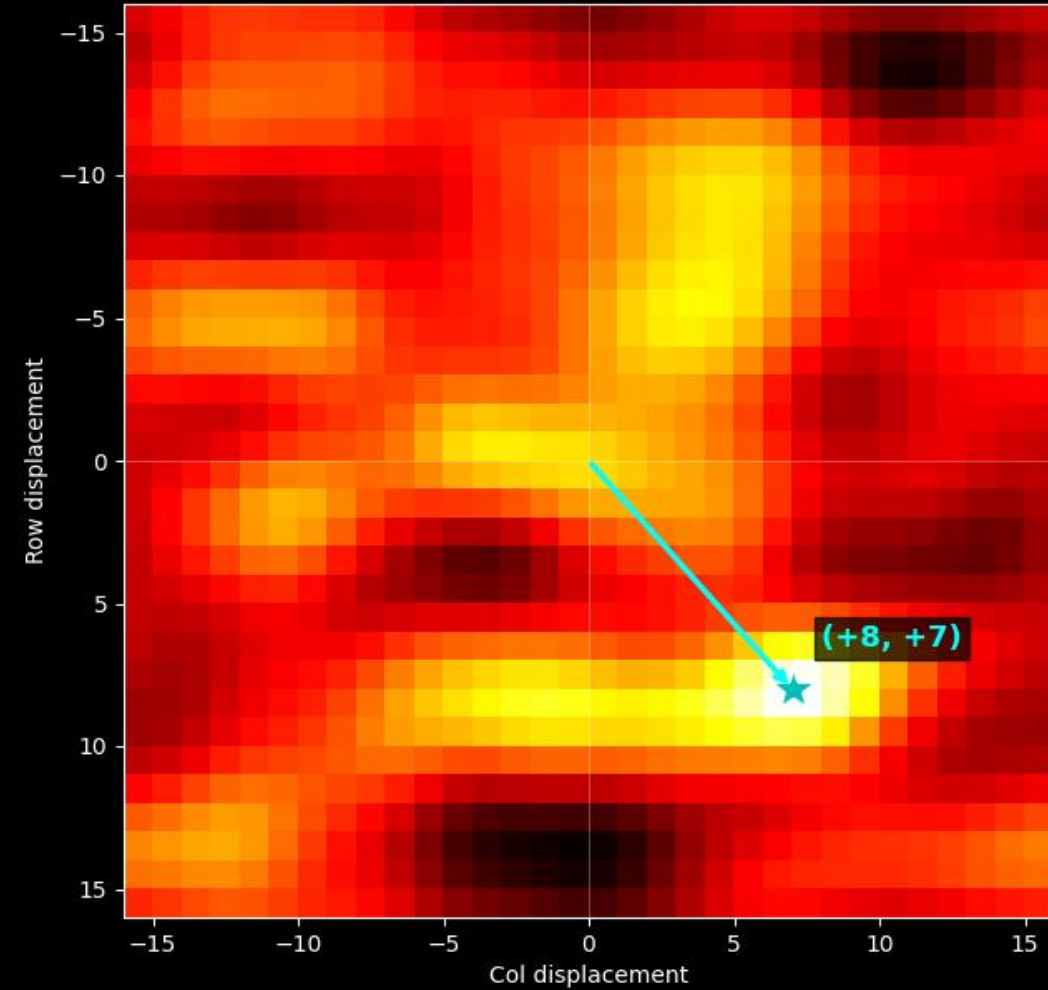
POC Demonstrations

Step 6 / 7 — KCF only: +7 px right, +8 px down (click for summary)

Frame 3 (Frame 3 (shifted)) | KCF ONLY | +7 px right, +8 px down



KCF Response | Peak at (+8, +7)



Domain Expert Review & Feedback

- **Name:** Mr. Senthilkumaran
- **Organization:** TVS Sundaram Industries Pvt Ltd, Defence Business Group
- **Expertise:** Engineering Leader in Defense and Aerospace sector with rich experience in R&D, complete product development for EV, Radar, Aerospace global customers
- **Key Suggestions:**
 - Full aerodynamic guidance is out of scope for a hardware PoC and distracts from the core processing goals.
 - Prioritize the critical "Sensor-to-Shooter" loop latency and accuracy.
 - Shift focus to demonstrating Shakti's capability for deterministic I/O in bounded timeframes.
- **Actions Taken / Planned:**
 - Transitioned entirely from drone guidance to a passive Infrared pattern recognition system.
 - Realign the project to target Counter-UAS and defence surveillance, validating Shakti's I/O reliability in real-world tactical scenarios.

MVP Panel Feedback & Improvements

Key Suggestions	Changes Implemented	Impact
1. GPS-denied navigation is an established, solved problem	1. Dropped aerodynamic guidance to build a passive Infrared Search and Track architecture for Counter-UAS.	1. Delivers an emissions-free, tracking solution immune to radar jamming and electronic warfare.
2. LWIR selection lacked strong tactical justification over standard IR.	2. Repositioned LWIR specifically for SWaP-C optimized tracking of airframe thermal contrast at tactical ranges.	2. Explicitly validates Shakti's bare-metal, jitter-free I/O reliability in mission-critical defence loops.
3. Autonomous missile guidance is risky; a Human-In-The-Loop approach is preferred.		3. Directly aligns with high-demand Anti-Terrorism Force Protection systems by solving edge-processing bottlenecks.
4. Practical differentiation from existing defence solutions was unclear.	3. FPGA performs parallel pixel processing; Shakti RISC-V handles deterministic actuator I/O interface.	

Execution Plan for Finals

- What will be demonstrated
 - Seamless integration of the camera sensor, FPGA processing pipeline, and Shakti control unit.
 - A clean physical casing for the final module.
- Key tasks to be completed:
 - Translate the validated Python KCF tracking model into optimized RTL/Verilog for the FPGA.
 - Fuse the Shakti processor with the custom KCF FPGA IP core.
 - SWaP-C Optimization: Design and fabricate a custom PCB to house the integrated architecture (if required).
 - Conduct rigorous latency benchmarking to maximize tracking accuracy and minimize the "Sensor-to-Shooter" response time.
- Implementation plan:
 - Shift the KCF matrix operations from Python validation to Verilog to achieve fast pattern match.
 - Upgrade the camera interface from UART to a high-speed parallel bus to drastically cut data ingestion latency (If possible with the current hardware constraints).
- Validation approach:
 - Benchmark lock-on and tracking stability using predefined static and moving reference targets.
 - Determine the maximum target velocity before lock-loss or tracking degradation occurs.
- Timeline
 - [30 April] KCF Verilog Porting & AXI Interface.
 - [30 April] Shakti RTOS application
 - [30 May] Shakti + FPGA IP Integration.
 - [30 August] Final Benchmarking & Module Casing.

Thank You